

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		to ensure that the concentration of AgNO ₃ in the burette is accurate / does not change			1	1		1
		(ii)		award (1) for any of following - undiluted seawater would give too high a titre / a titre of 132 cm³ - concentration of Cl⁻ ions is too high (in undiluted seawater) - reduces Cl⁻ ion concentration (so less AgNO₃ needed)			1	1		1
		(iii)		use a pipette/burette to measure seawater (1) into a volumetric flask (1) add 50 cm ³ of seawater to a 250 cm ³ volumetric flask (and make up to the mark with deionised water) / 20 cm ³ of seawater to a 100 cm ³ volumetric flask (and make up to the mark with deionised water) (1)	1 1		1	3	1	3
		(iv)		award (1) for any action and (1) for corresponding explanation e.g. add drop by drop ... to avoid 'overshooting' the endpoint shake the flask/rinse the side of the flask ... to ensure that all the reactants react place flask on a white tile ... to clearly see the colour change			2	2		2
		(v)		2Ag ⁺ (aq) + CrO ₄ ²⁻ (aq) → Ag ₂ CrO ₄ (s) ignore state symbols		1		1		